BGIN SR 00**
(Working Draft)

# Layer2 Governance Study Report

This study is a work product of IAM, Key Management and Privacy Working Group of BGIN.

©2025 BGIN (Blockchain Governance Initiative Network) is registered as BN Association, a Japanese general association with its principal place of business at Shiba Building, 704, 4-7-6, Shiba, Minato-ku, Tokyo. All right reserved.


© BGIN – All rights reserved

# Abstract


© BGIN – All rights reserved

# Introduction

Disclaimer: The views expressed in the document are personal views of the participating members of the BGIN community and should not be seen as the official views or recommendations of the institutions with which they are affiliated.

The technology described in this document was made available from contributions from various sources, including members of the BGIN and others. Although the BGIN has taken steps to help ensure that the technology is available for distribution, it takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this document or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any independent effort to identify any such rights. BGIN and the contributors to this document make no (and hereby expressly disclaim any) warranties (express, implied, or otherwise), including implied warranties of merchantability, non-infringement, fitness for a particular purpose, or title, related to this document, and the entire risk as to implementing this document is assumed by the implementer. The BGIN Intellectual Property Rights policy requires contributors to offer a patent promise not to assert certain patent claims against other contributors and against implementers. BGIN invites any interested party to bring to its attention any copyrights, patents, patent applications, or other proprietary rights that may cover technology that may be required to practice this document.


© BGIN – All rights reserved

# Table of contents

**Abstract** **2**
**Introduction** **3**
**Table of contents**

30

## 1. Introduction

The invention of Bitcoin evoked the possibility of achieving "decentralization" in financial transactions without the need for third-party intervention. Based on Bitcoin's fundamental principles, numerous communities and businesses have emerged with the aim of adding new functionalities beyond simple value exchanges while preserving 'decentralization.'

On the other hand, Bitcoin deliberately limits its functionality to asset transfers, maintaining an optimal level of "decentralization" through cryptographic and economic considerations. The addition of new functionalities, therefore, should be critically examined, as it may inevitably compromise the degree of "decentralization" that Bitcoin aspires to uphold. In response to this challenge, the industry has shown a tendency to introduce new mechanisms (such as incentive structures) to address the trust points - essentially, the centralization risks - that arise as a result of functional expansions.

While it may be premature to fully assess the extent to which 'decentralization' can be maintained through these innovations, identifying the potential trust points associated with the additional functionalities that are currently being widely adopted in the industry will be invaluable. This analysis will contribute to the identification of governance risks and inform discussions on the regulatory frameworks for service providers that claim to uphold 'decentralization.'

There are various additional functionalities that introduce trust points. However, as indicated in the title of this paper, the focus here is on Layer 2 solutions, which are employed to enhance the scalability and cost efficiency of blockchains, particularly Bitcoin and Ethereum. After redefining Layer 2 solutions in the next section, this paper will analyze the trust points introduced by their adoption and subsequently discuss the associated governance challenges.

## 2. Layer1 Governance


© BGIN – All rights reserved

Before discussing the governance challenges introduced by Layer 2 solutions, it is essential to first examine the current governance structures of Layer 1 blockchains that rely on Layer 2, as well as the ethos and goals upheld by their respective communities. Understanding these foundational aspects will provide a clearer perspective on how Layer 2 solutions interact with and potentially impact the governance of Layer 1 blockchains.

## 2-1. Comparison of Bitcoin and Ethereum Governance

### 2-1-1. Operating Organizations

Bitcoin: Absence of a Centralized Governing Body

Bitcoin is uniquely decentralized in that it lacks a formal governing entity. Unlike traditional financial systems or even many blockchain projects, Bitcoin does not have a foundation or centralized authority overseeing its development and governance. Instead, its development is driven by an open-source community where contributors voluntarily engage in protocol maintenance and improvement. Funding for Bitcoin development primarily comes from independent organizations such as Chaincode Labs, Brink, and Blockstream, which provide grants and sponsorships to developers (Narayanan et al., 2016).

The decentralized nature of Bitcoin's governance ensures resilience against external interference but presents challenges in coordinating long-term development. The absence of a centralized decision-making body makes protocol upgrades a slow and consensus-driven process.

Ethereum: Coordination through the Ethereum Foundation

Ethereum, by contrast, operates under a more structured governance model led by the Ethereum Foundation (EF). The EF functions as a non-profit organization responsible for funding research and development, organizing developer conferences, and guiding the network's long-term vision (Ethereum Foundation, 2023). While Ethereum governance is more structured than Bitcoin's, it remains open-source and relies on a combination of Ethereum Improvement Proposals (EIPs) and developer consensus.

### 2-1-2. Development Funding and Inventive Structure

Bitcoin: Challenges in Sustainable Funding

Bitcoin's funding model relies on voluntary contributions, grants, and corporate sponsorships. Unlike Ethereum, Bitcoin did not conduct an initial coin offering (ICO), nor does it have a mechanism where a portion of block rewards is


© BGIN – All rights reserved

allocated to developers. Instead, Bitcoin's mining rewards are exclusively directed to miners, while developers must seek external funding from independent sources (Antonopoulos, 2017).

This model ensures decentralized funding, preventing any single entity from exerting control over Bitcoin's development. However, it also creates sustainability challenges, as funding for research and maintenance is not guaranteed. The reliance on external sponsorships from companies such as Blockstream raises concerns about potential influence from corporate interests.

Ethereum: Structured Financial Incentives for Development

Ethereum's launch was accompanied by an ICO in 2014, raising approximately $18 million, which provided the initial capital to support protocol development (Ethereum Foundation, 2015). This funding has enabled the EF to provide grants to developers, researchers, and infrastructure projects.

Additionally, Ethereum's ecosystem fosters a more sustainable financial model:

1. DeFi projects, Layer 2 solutions, and smart contract platforms generate revenue, which contributes to development funding.
2. Ethereum Foundation, Gitcoin grants, and DAOs provide structured funding mechanisms to support open-source contributors.
3. Developers can monetize their work through smart contract applications, creating an ongoing incentive for protocol improvements.

Ethereum's incentive structure enables rapid innovation, but it also creates a system where financial interests may influence governance decisions, potentially favoring large stakeholders over the broader community.

**2-1-3. Development Process and Governance**

Bitcoin: Conservative and Community-Driven Consensus

Bitcoin follows a highly conservative development approach, prioritizing security and stability over rapid innovation. The Bitcoin Improvement Proposal (BIP) process requires broad consensus among node operators, developers, and miners before a change is implemented. As a result, upgrades take significant time and effort to be widely accepted (De Filippi & Wright, 2018).

For example, Segregated Witness (SegWit) and Taproot, two of Bitcoin's most significant upgrades, underwent extensive discussions and adoption processes


© BGIN – All rights reserved

before implementation. While this slow-moving governance ensures protocol stability, it also makes Bitcoin less adaptable to emerging technical challenges.

Ethereum: Agile Governance and Frequent Upgrades

Ethereum adopts a more flexible and progressive development model. The Ethereum Improvement Proposal (EIP) process enables frequent updates, and decision-making is driven by core developers in collaboration with the broader community (Buterin, 2021). Examples of major protocol upgrades include:

- EIP-1559 (2021): Implemented a new fee-burning mechanism.
- The Merge (2022): Transitioned Ethereum from Proof of Work (PoW) to Proof of Stake (PoS).
- Upcoming Proto-Danksharding (2024): Aims to reduce rollup transaction costs.

The strong influence of core developers and the Ethereum Foundation enables Ethereum to rapidly adapt to market needs. However, it also introduces "trust points", where certain developers or organizations wield significant influence over network upgrades.

**2-1-4. Proof of Work vs Proof of Stake**

Bitcoin: Proof of Work (PoW) as a Decentralization Mechanism

Bitcoin’s security and decentralization are ensured through Proof of Work (PoW), which requires miners to expend computational power to validate transactions. PoW is highly resistant to centralization, as it relies on a global network of miners competing for block rewards (Nakamoto, 2008). However, mining has become geographically concentrated, particularly in regions with access to cheap electricity, leading to some degree of centralization in mining pools.

Ethereum: Transition to Proof of Stake (PoS) for Efficiency

Ethereum transitioned from PoW to Proof of Stake (PoS) in 2022. PoS provides several benefits:

- Energy efficiency: PoS significantly reduces Ethereum’s energy consumption.
- Lower barriers to participation: More users can become validators without specialized hardware.
- Scalability improvements: PoS enables future optimizations like sharding, increasing transaction throughput.


© BGIN – All rights reserved

However, PoS introduces centralization risks, as staking power is often concentrated in large pools such as Lido, Coinbase, and Binance (Ethereum.org, 2023). This raises concerns about wealth-based governance and validator collusion.

### 2-1-5. Node Operation

Bitcoin: Lightweight Full Nodes for Accessibility

Running a Bitcoin full node requires moderate storage (~500GB) and standard hardware, allowing broad participation. This ensures that Bitcoin's rules are enforced by a large, independent node network, reinforcing decentralization (Antonopoulos, 2017).

Ethereum: Heavy Storage Requirements but Expanding Capabilities

Ethereum nodes require higher storage (~1TB+) due to its account-based model and smart contract execution. This has led to reliance on centralized RPC providers like Infura and Alchemy, which are used by many wallets and dApps. However, ongoing efforts to decentralize infrastructure, such as stateless clients and decentralized RPC networks, aim to mitigate this risk (Ethereum.org, 2023).

### 2-1-6. Protocol Simplicity vs Functional Scope

Bitcoin: A Secure, Minimalist Protocol

Bitcoin adheres to a strictly minimalistic design, focusing solely on secure, decentralized peer-to-peer transactions. This simplicity enhances security and resilience, reduces attack vectors, and limits the need for protocol changes.

Ethereum: A Versatile and Expanding Ecosystem

Ethereum is Turing-complete, allowing the execution of smart contracts, DeFi, and Layer 2 rollups. While this expands Ethereum's use cases, it also increases complexity, attack surfaces, and reliance on additional security mechanisms.

### 2-1-7. Monetary Policy and Token Supply

Bitcoin: Fixed Supply Model


© BGIN – All rights reserved

Bitcoin has a fixed cap of 21 million BTC, making it scarce and deflationary (Antonopoulos, 2017). The predictable issuance schedule (halving every ~4 years) supports Bitcoin's role as digital gold.

Ethereum: Adaptive Monetary Policy

Ethereum does not have a fixed supply but introduced EIP-1559, which burns a portion of transaction fees to reduce inflation (Buterin, 2021). Since Ethereum's PoS transition, it has at times become deflationary when network activity is high.

| | Bitcoin | Ethereum |
|---|---|---|
| Operating Organization | No central governing body, relies on community-driven governance | Governed by the Ethereum Foundation, which coordinates development |
| Development Funding | Funded by grants, donations, and sponsorships, but lacks a structured funding model | Initially funded through ICO ($18M), ongoing funding from EF, DAOs, and DeFi projects |
| Development Process | Conservative, slow-moving BIP process ensuring security and stability | Frequent upgrades through EIPs, rapid adaptability but central developer influence |
| Consensus Mechanism | Proof of Work (PoW), energy-intensive but highly decentralized | Transitioned to Proof of Stake (PoS), energy-efficient but staking centralization risks |
| Node Operation | Lightweight full nodes (~500GB), ensuring broad participation | Heavy storage requirements (~1TB+), reliance on centralized RPC providers |
| Protocol Simplicity | Minimalist design focused on peer-to-peer transactions | Turing-complete smart contract platform, supporting DeFi and Layer 2 solutions |
| Monetary Policy | Fixed supply of 21 million BTC, deflationary economic model | No fixed supply, EIP-1559 burns fees, PoS reduces inflationary pressure |

**(Table 1) Comparison of Bitcoin and Ethereum Governance Models**

### 2-2. Ethos and Goals of Bitcoin and Ethereum

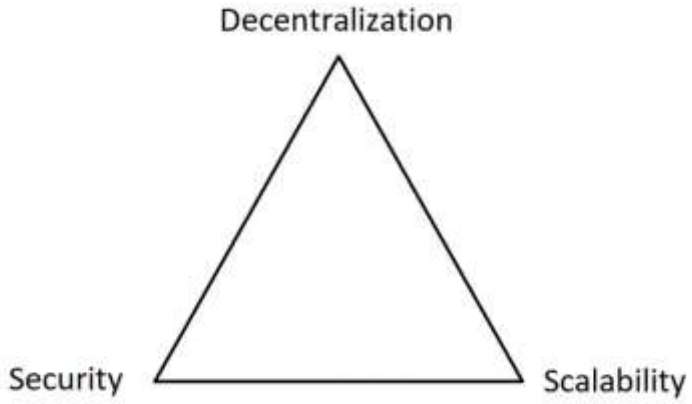
Bitcoin's development community prioritizes security and stability, favoring incremental upgrades over fundamental changes to its economic model or


© BGIN – All rights reserved

consensus mechanism. Rather than modifying the base layer, Bitcoin Core developers advocate for Layer 2 scaling solutions like the Lightning Network to enhance efficiency while preserving decentralization (Antonopoulos, 2017).

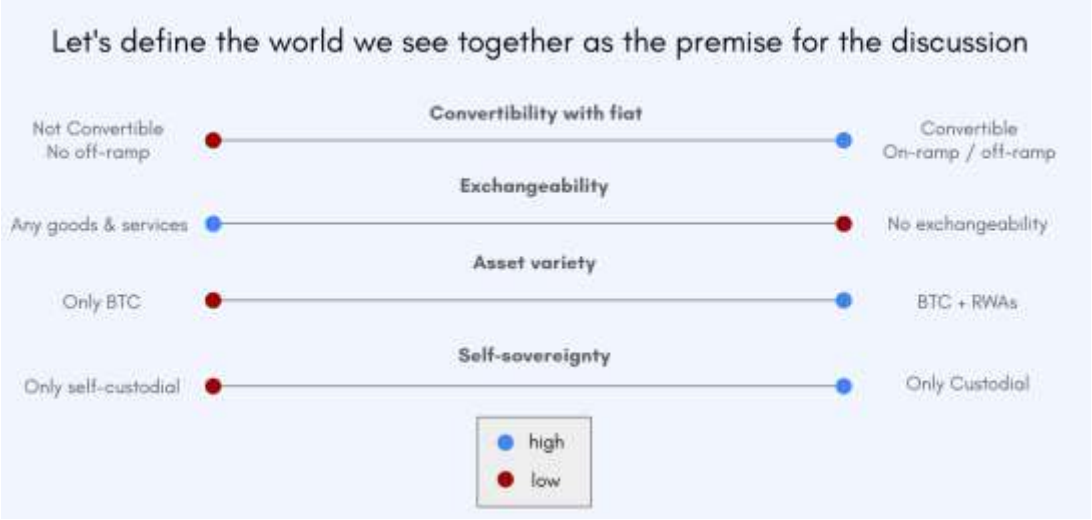
Ethereum, by contrast, embraces continuous innovation, positioning itself as a "world computer" that evolves to support decentralized applications (Buterin, 2014). The Ethereum Foundation promotes frequent upgrades, exemplified by the transition from Proof of Work (PoW) to Proof of Stake (PoS) and EIP-1559's fee-burning mechanism (Ethereum Foundation, 2023). Unlike Bitcoin's conservative approach, Ethereum follows a Layer 2-centric roadmap, shifting execution to rollups to improve scalability while maintaining decentralization (Ethereum Foundation, 2023).

While Bitcoin developers focus on preserving a stable store of value, Ethereum aims to be a flexible, scalable infrastructure for global computation and finance. These contrasting philosophies define the fundamental divide between the two networks. Given these differences, if we reference the Blockchain Trilemma of Security, Scalability, and Decentralization（Figure 1), we can say that while blockchain communities are striving to achieve all three, their priorities differ depending on their respective ethos and goals.




（Figure 1) Blockchain Trilemma (Vitalik Buterin, 2017)


© BGIN – All rights reserved




**(Figure 2) Key dimensions of Layer1 blockchain as asset transfer infrastructure**

In the discussion above, the ethos and goals of each community were examined. As an alternative perspective, when considering the role of blockchains as asset transfer infrastructure, it becomes evident that the ideal vision pursued by regulatory authorities often differs from that of Layer 1 blockchain communities. If this difference in perspective is not addressed, there is a risk of losing sight of the objectives of the discussion on regulations in Section 6: Discussion.

Therefore, to align perspectives, it is necessary to define the ideal vision that each blockchain community envisions when their blockchains are used as asset transfer infrastructure. The visions by Layer 1 blockchain communities introduced here are based on the framework used as a premise for discussing the regulatory challenges of the Lightning Network, which was conducted with Bitcoin Core developers at BGIN Layer2 Meetup @Tokyo in September 2024.

As illustrated in Figure 2, when considering the role of Layer 1 blockchains as asset transfer infrastructure, four key dimensions should be taken into account in descending order: convertibility with fiat currencies, exchangeability with goods and services, asset variety, and self-sovereignty.

The ultimate ideal world originally envisioned by Bitcoin Core may have been one where Bitcoin does not need to be convertible into fiat currencies, fully functions as a medium of exchange that can be used to trade for any goods or services, where BTC is the sole currency in use, and where individuals conduct peer-to-peer transactions exclusively through self-custodial wallets without involving any third parties.


© BGIN – All rights reserved

**(Figure 3) Definition of the ideal world where Layer 1 blockchains serve as asset transfer infrastructure for Layer 1 blockchain communities in this paper**

However, only a small fraction of people still adhere to this vision. Given the current state of development, as shown in Figure 3, the more pragmatic world envisioned by core developers of both Bitcoin and Ethereum appears to be one where they support convertibility with fiat currencies, function to some extent as a medium of exchange for goods and services, and accommodate various types of assets beyond just native tokens (BTC or ETH). The extent to whether different types of assets should exist on-chain may be a point of divergence between Bitcoin developers and the Ethereum Foundation. Additionally, in this envisioned world, a significant number of people would engage in peer-to-peer transactions using self-custodial wallets. (However, perhaps, and somewhat regrettably, the ideal world has not been clearly defined even within the communities.)

While opinions on this matter may vary, and room for debate remains, this paper adopts this definition as the ideal world envisioned by these Layer1 blockchain communities for future discussion.

In the discussion in Chapter 2, governance, ethos, and goals of Layer 1 blockchain were examined through a comparison of Bitcoin and Ethereum. However, this analysis should also help identify key aspects to consider when evaluating the governance of other Layer 1 blockchains. Please keep note that this paper does not seek to determine which approach is inherently better or worse.

Rather, the objective of this paper is to assess how Layer 2 solutions—currently being developed and utilized—can be structured in a way that meets public interest (regulatory) requirements without undermining the ultimate goals set by each Layer 1 blockchain. With this in mind, the paper aims to explore what measures can be


© BGIN – All rights reserved

maximally taken by development communities (or service providers) and regulators to ensure that Layer 2 solutions operate in alignment with these goals, minimizing governance and regulatory challenges.

3. Layer2 Overview

3-1. Definition of Layer2

Bitcoin's L2 solutions include a variety of options beyond the Lightning Network, such as Taro, Liquid Network, Stacks, Rootstock (RSK), RGB, and Mintlayer. In contrast, Ethereum has an even greater number of L2 solutions. Notable examples include OP Superchain, Arbitrum, Starknet, Base, BNB Chain, zkSync, and Polygon zkEVM.

Various terms are used to describe these L2 solutions, including state channels, rollups (optimistic rollup/zk rollup), Plasma, Validium, Optimum, and sidechains. However, the terminology surrounding "L2" itself, as well as the classification of L2 types, tends to be used without clear definitions. Therefore, this paper will begin by defining the relevant terminology.

As explained in the previous chapter, since the goals pursued by each community differ fundamentally, it is not meaningful to discuss Bitcoin and Ethereum L2 solutions simultaneously. Hence, the following discussion will focus solely on Ethereum's L2 solutions.

In the Ethereum ecosystem, there is an abundance of self-proclaimed "L2" solutions, and the definition of what should be considered "L2" remains ambiguous. However, many solutions referred to as "L2" tend to have characteristics that go beyond merely being scaling solutions, such as:

- Enabling on-ramp and off-ramp transactions directly on the "L2" without relying on L1.
- Allowing engineers to develop applications and protocols solely on the "L2" without dependence on L1.
- Possessing their own governance structures (e.g., "decentralized" organizations or centralized organizations).

This situation raises concerns that some solutions may be leveraging the Ethereum brand for marketing purposes, even though they are essentially creating more centralized L1-like systems.

L2 solutions eventually need to meet two essential sets of conditions:


© BGIN – All rights reserved

A. **Conditions to prevent the compromise of L1 governance**
B. **Conditions required by regulators for customer protection and AML/CFT compliance when L2 acts as an asset transfer infrastructure**

If condition (A) is not met—meaning the L2 compromises L1 governance—it could negatively impact the entire ecosystem. If condition (B) is not met, all regulated entities could be prohibited from using that L1 or L2, creating a risk of legal consequences for developers, including potential lawsuits.

In this paper, only those solutions that satisfy condition (A) will be referred to as L2. Under this premise, an L2 must meet the following conditions:

1. It addresses the challenges of high fees and slow transaction speeds on L1.
2. It enables 'off-L1 chain' transactions.
3. It reduces the number of transactions processed on L1.
4. It has mechanisms to ensure the integrity of all 'off-L1 chain' transactions until they are recorded on L1.
5. It provides users with the ability to forcibly withdraw their assets to L1 at any time, ensuring user control and asset security regardless of operator behavior.

To ensure that Ethereum's governance is not compromised, it is essential that users can ultimately rely on the security of Ethereum itself. For this reason, conditions (4) and (5) must be strictly satisfied.

If Ethereum considers "decentralization" as one of its core ethos, one might question whether the existence of L2 solutions with centralized elements contradicts this goal. However, if conditions (4) and (5) are firmly met, it could be argued that, despite the centralization and potential single points of failure in L2 solutions, it is ultimately the user's choice to use them. Therefore, decentralization was not included as a mandatory condition in this definition.

3-2. Classification

First, L2 solutions will be classified into two categories: **Blockchain-based** and **Non-Blockchain-based**.

Among the Non-Blockchain-based L2, the most representative type is the **state channel** model. However, since it is only used for specific niche use cases and is not mainstream for Ethereum L2, it will be omitted from this discussion.

Blockchain-based L2 can be divided into the following categories:


© BGIN – All rights reserved

- **Plasma**
- **zk-Rollup**
- **Optimistic Rollup**
- **Validium**
- **Optimum**

However, since **Plasma** is not currently considered mainstream, this paper will focus on the remaining four categories for classification purposes.

4. Trust Points in Layer2

5. XXX

| Component | Responsible Parties | Obligations |
|---|---|---|
| Data Storage | L2 Developers / DAO | - Ensure due diligence on DAC<br>- Design mechanisms for DAC to behave honestly |
| | Data Availability Committee | - Implement real-time user notifications<br>- Ensure accurate data and disclosure and persistence |
| Sequencer | L2 Developers / DAO | - Ensure due diligence on sequencers<br>- Design mechanisms for sequencers to behave honestly |
| | Sequencer Operators | - Operate sequencers transparently<br>- Ensure real-time user alerts |
| Consensus Mechanism | L2 Developers / DAO | - Design diverse validator sets<br>- Optimize slashing conditions and rewards<br>- Notify users of consensus disruptions |
| | Validator Operators | - Perform honest block validations<br>- Report validator downtime |
| Oracle | L2 Developers / DAO | - Ensure due-diligence on oracle providers<br>- Design multi-oracle configurations<br>- Provide alerts when risks are detected |
| | Oracle Providers | - Provide accurate and verifiable data<br>- Notify users and developers when data discrepancies occur |


© BGIN – All rights reserved

| | | |
|---|---|---|
| Bridge Contract | L2 Developers / DAO | - Monitor third-party performance<br>- Design secure bridge architectures<br>- Fix and update contracts<br>- Issue immediate user alerts |
| | Auditors | - Conduct smart contract audits<br>- Report identified vulnerabilities transparently |
| Forced Withdrawal / Escape Hatch | L2 Developers / DAO | - Implement robust forced withdrawal mechanisms<br>- Provide clear technical documentation<br>- Send real-time user alerts |
| | Wallet Providers | - Offer intuitive UI/UX for users<br>- Send real-time user alerts |
| | Data Availability Committee | - Notify users when data risks arise |

6. Case Studies

    - Examples of Governance Failures
    - Examples of Security Failures

7. Discussions

    - Discussion in the Ethereum Foundation
    - Implications to DeFi Regulations and Cybersecurity

8. Conclusions


© BGIN – All rights reserved

# Appendix A – Acknowledgement

(Informative)

## A.1 Editors and Co-editors

- Takaya Sugino
- Mitchell Travers

## A.2 Contributors

- BGIN IKP Working Group


© BGIN – All rights reserved

# Appendix B – Informative reference

(Informative)

•

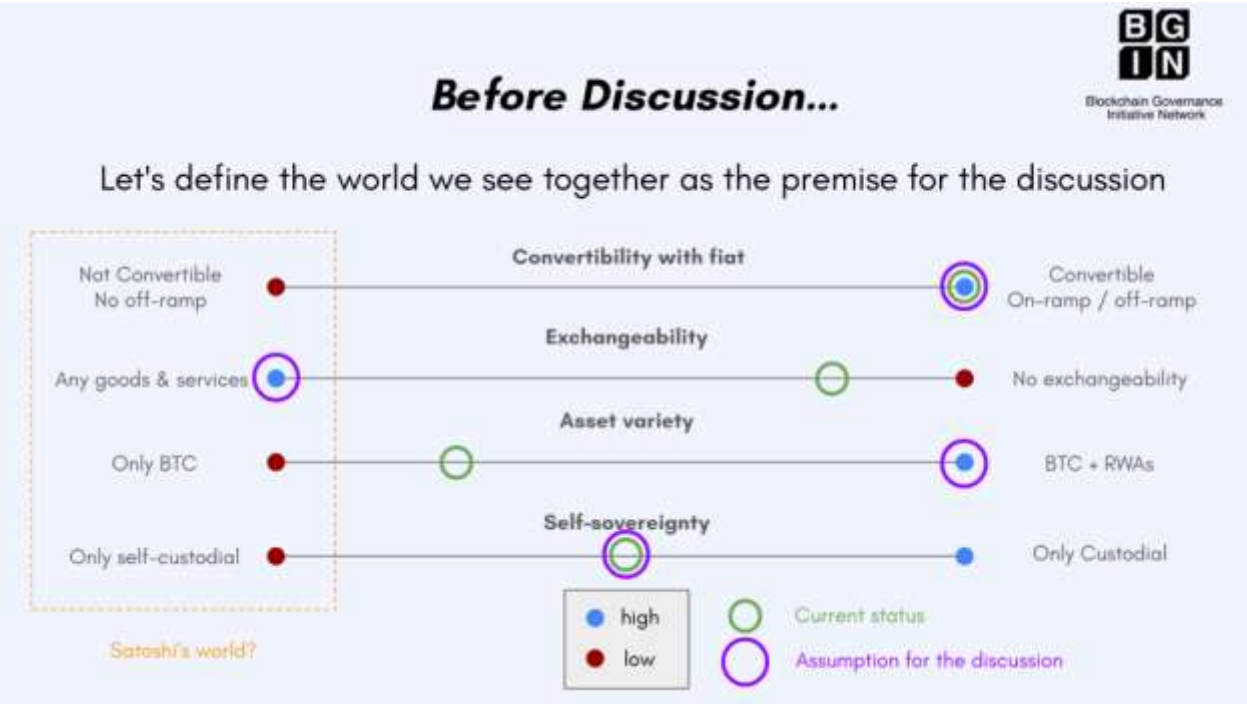

© BGIN – All rights reserved

# Key Artefacts:

- BGIN Layer2 Meetup@Tokyo Meeting Report

  240920_BGIN Layer2Meetup@Tokyo Meeting Report_Regulatory Uncertainty for Lightning Businesses

**Assumption for the discussion**




**Wallets**

- fake “self-custodial”

  Key questions

  - Is it a self-custodial wallet if you rely on third-party for running a lightning node?
  - What could the third party do if they run a lightning node for you?

  Discussion points

  - There’s a misconception that running a Lightning node for someone else can be done without full custody, but that’s not true. Companies might say they’re running the node for you while you maintain custody, but in reality, you’re trusting them with everything. They can create transactions that


© BGIN – All rights reserved

take all your funds, and you might unknowingly sign off on it. If you're not running your own node, and you're trusting a third party, you're essentially giving them full control over your funds.

- In Lightning, a third-party node operator can engage in several malicious activities. For instance, they can force-close channels or broadcast an outdated channel state to steal funds. Even though you might hold the private key to the Bitcoin network, the node operator still has a lot of control over the Lightning channels.
- The problem is that in Lightning, it's easy for a third party to trick you into signing something without fully understanding what it is. In theory, you should be able to sign transactions yourself, but in practice, you're often signing blindly, which can lead to losing funds if the third party is malicious.
- Users need to know what to sign and if you're not running the node yourself you don't know what to sign. Users also need to remember which revocation key to reveal.
- Many companies think they can have self-custody while outsourcing running their lightning node to another company. So far this isn't really possible technically. There are projects like VLS which may help but right now aren't really designed for third party nodes.
- The solution is to make it easier for individuals to run their own Lightning nodes, rather than relying on third parties. If people could run nodes on their phones or personal devices, they would have more control over their funds, and it would reduce the risks associated with trusting third-party node operators. Buy it's much easier and more tractable on layer 1. On layer 1, if you have a custodial wallet, the risks are more manageable. When sending Bitcoin, you don't need any external information; you know your UTXO, you can sign and create transactions offline, and then give that transaction to the provider. Hardware wallets, for example, handle this well. In Lightning, it's more complex. There are still risks with third parties on layer 1, but they are smaller and more understood by the industry.

- Security

  Key question


© BGIN – All rights reserved

- LN-supported wallets are mostly hot wallets. Do we need new operational requirements for custodial service providers managing LN-supported hot wallets?

Discussion points

- The "need to always watch the transactions on LN" is pretty well mitigated by watchtowers, which don't have private keys and can only defend channels from invalid closures. The main reason to keep the node online and key material in memory is because nodes need to sign to receive funds.
- Unlike layer one, in Lightning, you need to sign to receive payments. There's no passive reception. If you're routing funds, you need to be constantly online. If you go offline, you won't be able to receive, but your channels can still be safe if you have a watchtower.
- You often have to have your Lightning node online, meaning the key material is stored in RAM or on the device. Work is being done to move the key material to something more secure, like an HSM or a USB device, but right now, the private key is often actively used.
- Most exchanges would prefer cold storage, but they have to keep some funds online to manage Lightning transactions. Exchanges that implement Lightning have to manage the balance between security and functionality, often keeping some funds in hot wallets for Lightning operations.
- Some exchanges manage this by limiting the amount of money kept in Lightning nodes, replenishing from cold storage as needed. In the future, multi-signature solutions may help exchanges manage security better by splitting the keys across several servers, but this is still a few years away.
- A person I spoke to in the Bitcoin community said that Japanese exchanges hesitate to support Lightning due to regulatory uncertainties. But exchanges are concerned about securing customer assets, which is an important discussion point for implementing Lightning in Japan.

- KYC/ AML・CFT

Key questions

- How can we combat money laundering and other malicious activities?
- Is there any consideration of developing a solution to balance between them?


© BGIN – All rights reserved

- Any other measures being discussed in the Bitcoin community?

Discussion points

- In the Ethereum community, for example, Vitalik Buterin co-authored a paper that proposes using a mixing service to exclude bad actors while maintaining privacy, regardless of how effective and practical it is.
- The Lightning Network protocol developers are going in the opposite direction, enhancing privacy and making it harder to trace transactions. For example, if a company wants to know the origin of an HTLC, it can't because the protocol uses onion routing, and tracing the origin is difficult. Efforts are underway to decouple Lightning Network channels from the layer1 UTXOs, making it harder to link transactions. This may frustrate blockchain analytics companies, but the community is focused on privacy.
- While privacy is a priority, there isn't much interest in building solutions that help with AML/CFT measures. The focus is on making open-source software for individuals, and there's little concern for financial companies or exchanges.

**Routing (Intermediary) Node**

- Customer Asset Segregation

Key questions

- Can routing node operators lose deposited customer assets or engage in malicious activities with customer assets?

Discussion points

- Lightning routing nodes follow the paths directed by the network and have atomicity, so transactions either fail or succeed without intermediaries seizing assets. Also, as long as the routing node has enough liquidity, the system should work fine.
- If regulators tried to regulate these routing nodes, they'd probably just shut down. The Lightning protocol is designed so that routing nodes process payments without knowing where they come from or where they go. It's better to focus on endpoints and custodial services for regulation rather than routing nodes, which function similarly to miners on the Bitcoin network.


© BGIN – All rights reserved

    - The connection to routing nodes is usually automatic, and you connect based on liquidity and connectivity, not individual trust. The routing nodes have multi-sigs with other nodes, but typically, the routing node isn't directly connected to the customer. The routing node simply passes payments along. When you hold a self-custodial wallet and use Lightning, you need to connect to routing nodes with large liquidity. But you only need to have a direct channel with the routing node, and that's automatic. You don't usually deal directly with the routing node.
    - A routing node operated by a service provider has a multisig address with customers' nodes that have a direct channel with the routing node. But the service provider need their private key to move fund around, though the service provider can exit if the other is offline (which can be mitigated by watchtower)

- Limited transaction amount

  Key questions

    - Can the transaction amount be a reason not to be regualted?

  Discussion points

    - There are many bad activities possible with small transactions. And there are some technologies being developed to allow large transactions by splitting routing paths. Limited transaction amount itself could not be the reason not to be regulated.

- KYC and AML/CFT

  Key questions

    - Can KYC/AML・CFT measures be implemented in routing nodes?

  Discussion points

    - routing nodes can't link the sender and recipient because routing nodes can know only the nodes in front and behind and because the sender and recipient do not have their own multi-sigs when they do not have a direct channel.
    - Supposing that we can't make any changes to the privacy enhancing mechanisms of Lightning, routing nodes can see the persons in front and behind of the transaction path. If all intermediary nodes need to do KYC


© BGIN – All rights reserved

and due-diligence of the other party, we can link the sender and the recipient. But it costs too much and does not sound efficient.

- Decentralization

  Key questions

  - Can routing node owners engage in Intentional blocking if they are centralized?

  Discussion points

  - Technically possible by being offline for a long period. The routing node owner has bitcoin in multi-sigs with their customers. Customers can withdraw their assets after waiting for a certain period, when the routing node owner intentionally blocks transactions somehow.

  - If there is too much dependence on a few routing nodes, transaction failure and delay could happen easily. There is also a privacy risk that transaction analysis becomes easier. But they are not “regulatory” risks.

**Crypto Exchanges that support LN**

- Wallet management

  Key questions

  - It is not that they are required to hold customer assets in cold wallets (in Japan yes), but supporting LN makes them hold customer assets in hot wallets. Is this a security risk?

  Discussion points

  - See page 5

- KYC and AML/CFT

  Key questions

  - Is the due-diligence of the counterparty feasible? Custodial service providers might need to store past transaction data as a sender and receiver, but what about as an operator of a routing node?

  Discussion points


© BGIN – All rights reserved

- Existing crypto exchanges are required to perform due diligence on their counterparties and customers. When it comes to supporting Lightning, the exchange could potentially store past transaction commitment data and conduct KYC with anyone who has a direct channel to its node. While this is technically feasible, it becomes operationally complex and may impose limitations for blockchain analysis.

- Many exchanges think supporting Lightning is doable from a regulatory standpoint—it's not very different from Layer 1. But the big issue is that supporting Lightning requires significant engineering resources, and for most exchanges, it's not a huge moneymaker. While some of them might adopt Lightning due to customer demand, the complexity and cost may deter exchanges from pioneering this space.While regulation might seem like a barrier, it's more about the business case and engineering effort. It's a lot of work for exchanges to implement Lightning nodes securely.

**Watchtower**

Key questions

- Are there any evil acts that watchtowers can work on arbitrarily, while there are some functions to prevent evil acts. Watchtowers just have encrypted commitment transaction data. Only when it detects an attempt to broadcast an outdated commitment transaction, they can submit just the appropriate penalty transaction.

- How are watchtowers incentivized to act correctly?

- Is watchtower decentralized? Is there any single point of failure?

Discussion points

- The potential for collusion is minimal. Watchtowers can only fail by going offline. The data given to them is encrypted commitment transaction data and doesn't mean anything on its own. Only when it detects an attempt to broadcast an outdated commitment transaction, they can submit just the appropriate penalty transaction. There's no real mechanism for watchtowers to act maliciously, except by not performing their duty.

- Incentives for watchtowers are not well-defined since the need to act rarely occurs, making it hard to implement a meaningful reward system. The cost of running a watchtower is low, and services might charge a


© BGIN – All rights reserved

small monthly fee, but the current setup seems to work without extensive incentive mechanisms.

- Using multiple watchtowers to protect the same channel is allowed, so any individual tower is a single point of failure, but they can be redundant. Even if several try to act simultaneously, it doesn't break the system.

- ## BGIN Block#11Meeting Report

BGIN Block#11 MR L2 Governance

Layer2 Governance

Date: Monday, October 21, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

**Introduction and Setup**

- **Starting and Setup Issues:** The presentation is set to begin after some initial setup delays. Discussions on screen-sharing and technical issues are mentioned.
- **Casual Session Setup:** Small audience size, allowing for a more casual and interactive presentation style.

---

**Layer 2 Governance and Definitions**

- **Definition Ambiguity of L2:** The definition of Layer 2 (L2) solutions is ambiguous. Historically, L2 was viewed as a scaling solution for Layer 1 (L1) challenges like high fees and slow transaction speeds by handling transactions off the L1 chain.
- **Challenges with L2 Definition:** L2 solutions were intended to reduce the volume of L1 transactions by only recording final states. The definition of "off-chain" is also ambiguous and needs to be clarified as “off L1 chain.”
- **Potential Marketing Misuse:** "L2" has become a marketing term. Many projects call themselves L2 if they provide faster transactions and lower fees than L1, often without meeting core criteria.


© BGIN – All rights reserved

---

### L2 Characteristics and Scaling

- **Traditional L2 Characteristics:** L2s are expected to facilitate transactions faster and cheaper than L1, support the same virtual machine as L1 for application compatibility, and store a summarized version of L1 transactions.
- **Scaling and On/Off Ramp Transactions:** L2 solutions allow for direct on- and off-ramp transactions (fiat and crypto exchanges), bypassing L1. This raises questions about whether these are truly L1 scaling solutions.
- **L2 Development Freedom:** Engineers can develop applications on L2 using the same language and tools as L1, allowing for smooth interoperability.

---

### Governance in L2

- **Independent Governance:** L2s often have their own governance, sometimes through decentralized autonomous organizations (DAOs), using native or governance tokens. This structure is similar to L1 governance.
- **Centralization Concerns:** There is ongoing debate if such self-governed L2s are simply a form of L1. Their self-identification as L2 may be a strategy to imply they offer L1-level security.

---

### Understanding Sequencers and Data Storage in L2

- **Sequencers in L2:** Sequencers are responsible for validating transactions on L2 and submitting proof to L1. This role is essential for maintaining integrity between L2 and L1.
- **Data Storage and Cost Challenges:** Storing all L2 transaction data on L1 is costly due to limited L1 capacity. To address this, some L2 solutions use off-chain data storage layers for data availability without burdening L1 capacity.
- **Regulatory Touchpoints:** Sequencing and data storage are critical regulatory touchpoints. If L2 transaction data is hidden, users lose the ability to validate transactions, leading to potential issues with integrity and compliance.

---

### Key Concepts and Processes in Sequencing


© BGIN – All rights reserved

1. **Entering Phase:** Users lock assets in L1, and sequencers enable equivalent assets on L2 for use.
2. **Usage Phase:** Sequencers create a state root of L2 transactions and submit it to a roll-up contract on L1, frequently updating the state.
3. **Exiting Phase:** When users want to withdraw, they create a Merkle proof to confirm their transaction history on L2 before unlocking assets on L1.

---

**Proof Mechanisms: Fraud Proof vs. Validity Proof**

- **Fraud Proof:**
  - Users submit a Merkle proof, and challengers can submit a fraud proof if they detect discrepancies.
  - Fraud proof requires a challenge period where finality is delayed to allow challenges.
- **Validity Proof:**
  - Sequencers create zk-proofs that validate L2 transactions on L1, removing the need for user-submitted fraud proofs.
  - This method is more efficient and provides continuous validation.

---

**Risks in L2 Sequencing and Governance**

- **Smart Contract Vulnerabilities:** Smart contracts used for sequencing could fail or be hacked, posing risks for both fraud and validity proofs.
- **Centralization of Sequencers:** A single sequencer creates a central point of failure. Efforts in the Ethereum community aim to increase the number of sequencers for decentralized governance.
- **Challenge Period and Finality Risk:** In fraud-proof systems, users must wait for transaction finality until the challenge period ends. If sequencers misbehave and no one notices, users risk losing assets.
- **Deposit Incentive Risks:** Users who submit false transactions can lose their deposits if successfully challenged. A well-incentivized system is needed to discourage malicious behavior.

**Data Storage and Efficiency**

- **Call Data and EIP-4844**:
  - L2 transaction data stored in L1 using "call data" is more efficient than other methods but still costly.


© BGIN – All rights reserved

    - EIP-4844 introduces "blob data" for temporary, efficient storage, allowing better scalability for L2 solutions.
- **Data Availability Layers and Governance**:
    - These layers store L2 transaction data off L1, but must be monitored to avoid data tampering.
    - Governance tokens are sometimes used to incentivize accurate data monitoring, with penalties for misbehavior.

---

**Risks and Regulatory Considerations**

- **Smart Contracts and Sequencer Centralization**:
    - Both sequencer failure and smart contract vulnerabilities remain critical risk factors.
    - Data availability layers and governance mechanisms help mitigate some risks but introduce centralization concerns.
- **Fraud Proofs vs. Validity Proofs**:
    - Fraud proofs rely on challenge periods, where users can challenge invalid transactions.
    - Validity proofs, often utilizing zk-proofs, provide immediate validation but require high computational resources.

---

**Staking and Validator Risks**

- **Delegation Risks**:
    - Validators receiving delegated assets face unique regulatory and risk concerns. If slashing occurs, delegated assets might be lost, raising the need for regulatory oversight.
    - Pure staking with one's own tokens differs from delegated staking, as the latter involves handling others' assets and potentially greater regulatory touchpoints.

---

**Money Laundering Risks in Layer 2 and Lightning Networks**

- **Money Laundering through L2s**:
    - Repeated entry and exit from L2s could obscure transaction origins, complicating monitoring efforts.


© BGIN – All rights reserved

    - Lightning Network transactions are harder to trace due to privacy enhancements, presenting unique challenges for law enforcement.

---

**Optimistic Rollups and Security**

- **Challenge System in Optimistic Rollups**:
    - Users can submit Merkle proofs as challenges if they suspect a sequencer has submitted invalid data.
    - Challenges are verified via smart contracts, and penalties apply for invalid challenges, promoting secure and transparent operations.

---

**Open Questions and Final Thoughts**

- **Centralized Sequencers and Finality**:
    - L2 solutions might still rely on L1 for finality, affecting their independence as truly decentralized solutions.
    - There's discussion on how to balance efficient processing with decentralized validation and security.
- **Engagement with the Ethereum Community**:
    - Future discussions will explore solutions for the identified risks, particularly with ongoing developments in Ethereum's community, such as proposer-builder separation.

---

**Next Steps**

- **Gathering Community Feedback**:
    - A proposal to solicit public contributions, particularly from the Ethereum community, aims to refine risk analyses and possible solutions.

# Links and References


- Antonopoulos, A. (2017). *Mastering Bitcoin*. O'Reilly Media.
- Buterin, V. (2021). *Ethereum: A next-generation smart contract and decentralized application platform*.
- De Filippi, P., & Wright, A. (2018). *Blockchain and the Law: The Rule of Code*. Harvard University Press.



© BGIN – All rights reserved

- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.
- Ethereum.org. (2023). *Ethereum documentation*.



© BGIN – All rights reserved